

The Juggler Map and the $3n \pm 1$ Maps

Exact Coding and Arithmetic Obstructions

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Abstract

The Juggler map and the shortcut $3n + 1$ and $3n - 1$ maps share the word multiplier $3^o/2^L$, but realize their words in different arithmetic and metric spaces. Classical parity coding defines an exact map from every Juggler orbit into the 2-adic $3n - 1$ system. On actual periodic Juggler orbits it preserves every return time, and its value is an ordinary integer exactly when an explicit word divisibility holds. A family of genuine Juggler prefixes shows that arbitrarily precise modular return does not force that divisibility. We count this constructed family with an exact leading constant and prescribe its residues throughout the final even run. For the fixed word OOE, we also give an explicit counting error and a uniform polynomial first-witness bound in the modulus; this quantitative extension has a complete Lean proof, with independent review pending. On the positive integers, we prove that every $3n - 1$ target prime to three has at least $X^{21/25}$ ancestors below X , for all sufficiently large X . This signed adaptation uses height-corrected inverse-tree inequalities, a closed root domain, and an exact integer certificate with 177147 rows. A mean inequality places every positive certificate in the fixed $1/50$ grid, for either sign and at every finite residue level, strictly below exponent 0.99 in the at-most-linear range. Lean formalizations accompany the orbit, period, counting, grid-ceiling, and modular-return results, including the latter's analytic recurrence input. These results separate symbolic correspondence from integer realization and reciprocal-mass transport; no universal termination or unrestricted cycle exclusion theorem is asserted.

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1. Introduction

We compare the maps

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd,} \end{cases} \quad C_\sigma(x) = \begin{cases} x/2, & x \text{ even,} \\ (3x + \sigma)/2, & x \text{ odd,} \end{cases} \quad \sigma \in \{-1, +1\}.$$

Unless a larger domain is specified, inputs are positive integers. Write $g = C_{-1}$. On ordinary integers $C_{+1}(-x) = -g(x)$; positive $3n - 1$ orbits correspond to negative $3n + 1$ orbits, not to positive $3n + 1$ orbits.

Along a prescribed branch word, the leading Collatz multiplier acts on x , whereas the leading Juggler multiplier acts on $\log n$. The same letters generate the same formal walk one logarithm

apart. Turning this observation into a statement about integer trajectories requires additional information. This paper asks which information survives the correspondence.

Sections 2 and 3 construct the exact code and prove period preservation. Section 4 exhibits actual modular returns whose associated periodic-word codes are nonintegral. Section 5 proves a signed ancestor-count theorem, including the height correction that prevents an automatic transfer of the positive-map proof. Section 6 proves a limitation of the fixed grid used in that argument. Section 7 distinguishes counting from reciprocal mass and bounded word identities from unbounded stopping identities.

1.1. Contributions and prior work

Terras [T76] supplies the classical parity-vector bijection and stopping-word combinatorics. Bernstein and Lagarias [BL96] supply the 2-adic inverse parity map, including the signed affine family. We use that construction, rather than claim a new coding mechanism. Prasad and Prasad [PP25] discuss the leading-word approximation for Juggler-like sequences. The shared multiplier and the word counts are background.

The Juggler-specific refinement in Section 3 is preservation of every return time on an actual periodic orbit. Its proof combines classical coding with monotonicity within each branch. The integrality criterion is the classical affine cycle equation stated alongside this refinement. Section 4 gives a quantified obstruction involving actual floor iterates. Its counting and residue corollaries extract further consequences of classical power equidistribution; no new general distribution criterion is claimed. Theorem 4.4 adds a quantitative specialization for the single word OOE.

Krasikov and Lagarias [KL03, Theorem 6.1] prove an ancestor exponent 0.84 for every positive target prime to three under C_{+1} . Their theorem is not a statement about positive g -targets. Our Section 5 proves the latter statement by a strict-grid construction. The 1/50 cap table and induction measure follow M. Sharpe’s formal implementation [S26], with attribution and its MIT notice retained in the source. The signed height estimates, the domain for arbitrary positive unit targets, and the independently generated certificate are supplied here. We claim neither priority for the grid method nor a larger positive-map exponent.

Section 6 isolates a common mean obstruction for this fixed grid at every finite residue level. This is a limitation of those certificate inequalities, not a bound on actual ancestor growth.

The companion papers [A], [B], [C], and [D] treat Juggler cycle bounds, finite-depth descent statistics, fate contagion, and bounded-run $3n - 1$ cycles. We cite their results without reissuing their proofs. The present main theorems require neither their large computational floors nor the analytic estimates in [B].

1.2. Proof status

This is the living preprint, version 0.7.1. It has not been deposited or independently refereed. Mathematical priority for the signed adaptation and the isolated obstruction results remains subject to specialist review.

Theorems 2.1, 3.2, 4.1, 4.4, 5.1, and 6.1 and Corollaries 4.2-4.3 have compiled Lean statements; Appendix B identifies their precise scope. Theorem 4.1 is unconditional in Lean: first-derivative estimates, mixed-power cancellation, and the Fourier criterion prove its simultaneous-box recurrence input. Theorem 4.4 also has a complete quantitative Lean proof, including the finite derivative tests, every cutoff Fourier mode, the half-open box estimate, both counting errors, and the bounded witness. Appendix C gives an alternative written argument using [AR24]; the formal proof independently derives sufficient third- and fifth-derivative estimates and does not import the exact external formula. The finite certificate is checked with exact integers both in Lean and by an independent Python verifier. The numerical search that found its weights

is outside the proof. Kernel checking, agreement between prose and formal statements, and independent mathematical review are distinct checks.

2. Words and the exact orbit code

For $w = (b_0, \dots, b_{L-1}) \in \{0, 1\}^L$, with 0 denoting E and 1 denoting O, write $o(w) = \sum_i b_i$ and

$$\rho(w) = \frac{3^{o(w)}}{2^L}, \quad A(w) = \sum_{i=0}^{L-1} b_i 2^i 3^{\sum_{j>i} b_j}. \quad (2.1)$$

Induction on the prescribed affine branches gives

$$2^L C_{\sigma, w}(x) = 3^{o(w)} x + \sigma A(w). \quad (2.2)$$

This describes an actual orbit only when all branch guards hold. For an actual Juggler itinerary w starting at n , the inequalities $J(n)^2 \leq n$ on E and $J(n)^2 \leq n^3$ on O give

$$(J^L(n))^{2^L} \leq n^{3^{o(w)}}. \quad (2.3)$$

Thus a contracting prefix forces descent for $n > 1$. For positive $3n + 1$, the nonnegative correction in (2.2) instead prevents descent along a prefix with $\rho \geq 1$. Shared combinatorics does not remove this sign difference.

Let $d_0 < d_1 < \dots$ be the odd times of the Juggler orbit of n , finite or infinite, and define

$$H(n) = \sum_j \frac{2^{d_j}}{3^{j+1}} \quad \text{in } \mathbb{Z}_2. \quad (2.4)$$

An empty sum is zero. The summands' valuations tend to infinity, so the sum converges 2-adically. Both C_σ extend to $\mathbb{Z}_2$, using the residue modulo two to select the branch. Formula (2.4) is a specialization of the inverse parity map of [BL96].

Theorem 2.1 (exact signed orbit code). For every positive integer n ,

$$H(n) \equiv n \pmod{2}, \quad H(J(n)) = g(H(n)), \quad -H(J(n)) = C_{+1}(-H(n)). \quad (2.5)$$

Two starts have equal codes if and only if all their finite Juggler parity itineraries agree. On every finite source set, counting a prescribed length- d itinerary is exactly counting its associated code residue modulo 2^d .

Proof. Deleting an even first letter gives $H(n) = 2H(J(n))$. Deleting an odd first letter gives $H(n) = 1/3 + (2/3)H(J(n))$. These identities give parity and the minus-step identity. Negation gives the plus-step identity. Iterate. The classical parity bijection identifies length- d words with residues modulo 2^d . Equality of codes therefore gives equality of every prefix; conversely, equality of all residues gives equality in $\mathbb{Z}_2$. The finite-set assertion is equality of its membership conditions. $\square$

The formal construction takes compatible finite residues and their 2-adic limit. Its equality with (2.4) is proved by a telescoping sum whose remainder after time k has 2-adic norm at most 2^{-k} . The formal odd-time index uses the number of earlier odd times as j ; finite and empty odd-time families are included. This assumes neither termination nor 2-adic continuity of Juggler on its original integer inputs.

Example 2.2 (rational values and collisions). From $H(1) = 1$, following 3, 5, 11, 36, 6, 2, 1 backwards gives

n	3	5	11	36	6	2	1
$H(n)$	83/27	37/9	17/3	8	4	2	1.

Also $H(4) = H(6) = 4$. Thus the code neither takes every positive integer to an ordinary integer nor is injective. Every terminating start has a positive rational code whose reduced denominator divides a power of three, by backwards application of $q \mapsto 2q$ and $q \mapsto (2q + 1)/3$ from 1.

Corollary 2.3 (the distribution question is unchanged). Frequencies 2^{-d} for all length- d Juggler words, for every fixed d , are equivalent to

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \{1 \leq n \leq N : H(n) \equiv r \pmod{2^d}\} = 2^{-d}$$

for every d and code residue r . This equivalence proves neither condition. The formal statement transfers limits using exact equality of the finite-source counts.

3. Actual periods and ordinary integrality

Lemma 3.1 (order within a code fibre). If $H(n) = H(m)$ and $n \leq m$, then $J^k(n) \leq J^k(m)$ for every $k \geq 0$.

Proof. Equal codes imply equal parity at each time. Each branch function is nondecreasing. Induction applies the same branch to the two current values. $\square$

Theorem 3.2 (period preservation and integrality). Suppose $J^L(n) = n$ with $L > 0$, and let w be that actual itinerary. Put $D = 3^{o(w)} - 2^L$. Then $D \neq 0$, and:

1. $H(n) = A(w)/D$ in the 2-adic field.
2. For every $d \geq 0$, $g^d(H(n)) = H(n)$ if and only if $J^d(n) = n$. The analogous statement holds for C_{+1} at $-H(n)$.
3. The least periods of these three points are equal.
4. $H(n)$ is an embedded ordinary integer exactly when $D \mid A(w)$. In that case its ordinary signed Collatz orbit has the same return times as the Juggler orbit.

Proof. Iteration of (2.5) gives

$$3^{o(w)} H(n) = 2^L H(J^L(n)) + A(w).$$

The odd integer D is nonzero and invertible in $\mathbb{Z}_2$. Closing the orbit gives the formula.

One implication about return times is (2.5). Conversely, if the code returns after d steps, J^d preserves its fibre and is nondecreasing there by Lemma 3.1. If $J^d(n) > n$, every subsequent application stays strictly above n ; if $J^d(n) < n$, every subsequent application stays strictly below n . Both contradict $(J^d)^L(n) = n$. Hence $J^d(n) = n$. Equality of return times gives equality of least periods.

Finally A/D is an ordinary integer exactly when $D \mid A$. Injectivity of the rational embedding into the 2-adic field makes this necessary as well as sufficient. The integer maps agree with their 2-adic extensions on embedded integers. $\square$

The periodicity assumption on the Juggler start is essential. A periodic code alone has not been shown to imply a periodic start. For a nontrivial Juggler cycle, $D > 0$ by strict power-envelope expansion [A]. Its code is a positive rational g -periodic point, and its negative is a negative C_{+1} -periodic point. No result here establishes $D \mid A(w)$ for such a cycle.

4. Modular return does not force integrality

Theorem 4.1 (actual modular returns with large code denominator). Fix $a, b \geq 1$ with $3^a > 2^{a+b}$. For every integer $M \geq 1$ and lower bound B , infinitely many odd $n > B$ have actual itinerary $O^a E^b$, with

$$J^j(n) \geq n \quad (0 \leq j \leq a+b), \quad J^{a+b}(n) > n, \quad J^{a+b}(n) \equiv n \equiv 1 \pmod{2M}. \quad (4.1)$$

The rational periodic code associated with this word has denominator

$$q_{a,b} = \frac{3^a - 2^{a+b}}{\gcd(3^a - 2^a, 2^b - 1)}. \quad (4.2)$$

These denominators are unbounded at fixed b . In particular M may be any fixed power of $q_{a,b}$.

Proof. Put $s = 1 + 2Mt$, $n = s^{2^{a-1}}$, and $\beta = 3^a/2^{b+1}$. The first $a - 1$ odd steps are exact:

$$J^i(n) = s^{3^i 2^{a-1-i}} \quad (0 \leq i < a).$$

The next value is $y_0 = \lfloor s^{3^a/2} \rfloor$. Whenever the subsequent sources are even, their values are $y_j = \lfloor s^{2^{b-j}\beta} \rfloor$, $0 \leq j \leq b$. Nested square-root floors are exact, since $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$.

As t varies, the vector

$$\left(\frac{s^\beta}{2M}, \frac{s^{2\beta}}{2}, \frac{s^{4\beta}}{2}, \dots, \frac{s^{2^b\beta}}{2} \right) \pmod{1} \quad (4.3)$$

is uniformly distributed. Every nonzero integer linear combination has a nonintegral leading exponent: all exponents have odd numerator and denominator divisible by two. It is a Hardy-field function of polynomial growth, and its distance from any rational polynomial, divided by $\log t$, tends to infinity. Boshernitzan's criterion [Bo94, Theorem 1.3], also stated in [R26, Theorem 1.1], applies. Weyl's criterion gives joint distribution.

Place the first coordinate's fractional part in $[1/(2M), 2/(2M))$, and the others in $[0, 1/2)$. The box has volume $1/(2^{b+1}M)$. It enforces $y_b \equiv 1 \pmod{2M}$ and even $y_0, \dots, y_{b-1}$, so the word is actual. Since $\beta > 2^{a-1}$, its endpoint exceeds n for sufficiently large s . The odd portion increases and the even portion decreases to that endpoint, proving (4.1).

For $O^a E^b$, (2.1) gives $A = 3^a - 2^a$ and $D = 3^a - 2^{a+b}$. The oddness of A gives $\gcd(A, D) = \gcd(A, 2^b - 1)$, proving (4.2). The quotient is at least $(3^a - 2^{a+b})/(2^b - 1)$, which tends to infinity with a at fixed b . $\square$

The formal construction proves the exact root-floor identities, all branch guards, and every conclusion of (4.1) for each successful parameter with $s \geq 2^{2^{b+1}}$. It also proves (4.2) and, for every Q , the explicit sufficient bound

$$a \geq 2(2^b + (Q + 1)(2^b - 1)) \implies q_{a,b} > Q.$$

The formal analytic proof establishes BoxRecurrence: for every T there is a parameter $t \geq T$ in the simultaneous box above. It uses an elementary derivative argument in place of the Hardy-field criterion. For a phase with leading term cx^p , $c \neq 0$, $0 < p < 1$, the first-derivative estimate gives cancellation. A mean-value point in each fixed shifted interval proves that the actual difference lowers the leading exponent by one; induction and van der Corput's inequality handle higher noninteger powers. The same argument controls lower-order terms and the progression $s = 1 + 2Mt$. Uniform density of Fourier monomials on the torus, followed by a nonnegative continuous function supported in the interior of the target box, supplies arbitrarily late visits. Thus PaperERecurrence.theorem41 has no recurrence or exponential-sum hypothesis.

The theorem concerns growing prefixes and their associated *periodic-word* codes. It does not identify that code with $H(n)$, which depends on the entire subsequent itinerary. It gives no common start at all precisions, no actual nontrivial cycle, and no denominator bound for true Juggler cycles. Finite modular return is not exact orbit closure.

Corollary 4.2 (counting the constructed returns). Fix a, b, M as in Theorem 4.1 and put $d = 2^{a-1}$. Let $\mathcal{T}$ be the set of nonnegative parameters t for which (4.3) lies in the half-open box used above and $s = 1 + 2Mt \geq 2^{2^{b+1}}$. Define

$$R_{a,b,M}(X) = \#\{t \in \mathcal{T} : (1 + 2Mt)^d \leq X\}.$$

Then, as $X \rightarrow \infty$,

$$R_{a,b,M}(X) \sim \frac{X^{1/d}}{2^{b+2}M^2}. \quad (4.4)$$

For every fixed $\varepsilon > 0$, every sufficiently large interval $(X, (1 + \varepsilon)X]$ contains a start from this family and hence an actual modular return satisfying (4.1).

Proof. Uniform distribution and the box volume give

$$\frac{\#(\mathcal{T} \cap \{0, \dots, N-1\})}{N} \rightarrow \delta := \frac{1}{2^{b+1}M}.$$

The threshold excludes only finitely many parameters. For $X \geq 1$, the number of possible parameter positions is exactly

$$N(X) = \left\lfloor \frac{X^{1/d} - 1}{2M} \right\rfloor + 1 \sim \frac{X^{1/d}}{2M}.$$

Distinct parameters give distinct starts. Substituting $N(X)$ into the density limit yields (4.4). The difference $R_{a,b,M}((1 + \varepsilon)X) - R_{a,b,M}(X)$, divided by $X^{1/d}$, tends to $((1 + \varepsilon)^{1/d} - 1)/(2^{b+2}M^2) > 0$. It is therefore positive for every sufficiently large X . $\square$

This counts the explicit perfect-power family, not all starts satisfying (4.1). In particular, it is not a positive-density assertion among all positive integers.

Corollary 4.3 (prescribed residues in the even run). Fix the same a, b, M , and choose residues $0 \leq r_j < 2M$, $0 \leq j \leq b$, with r_j even for $j < b$ and $r_b = 1$. Infinitely many starts $n = (1 + 2Mt)^{2^{a-1}}$ satisfy (4.1) and

$$J^{a+j}(n) \equiv r_j \pmod{2M} \quad (0 \leq j \leq b). \quad (4.5)$$

More precisely, the constructing parameters beyond the same threshold, defined by

$$\left\{ \frac{s^{3^a/2^{j+1}}}{2M} \right\} \in \left[\frac{r_j}{2M}, \frac{r_j + 1}{2M} \right) \quad (0 \leq j \leq b), \quad (4.6)$$

have natural density $(2M)^{-(b+1)}$ in the parameter t .

Proof. The distinct positive noninteger exponents are unchanged; scaling every coordinate by $1/(2M)$ preserves the nonzero-mode argument. Joint uniform distribution gives the product of the $b + 1$ interval lengths, hence the asserted density. For every nonnegative real x , the fractional-part condition $\{x/(2M)\} \in [r/(2M), (r + 1)/(2M))$ is equivalent to $\lfloor x \rfloor \equiv r \pmod{2M}$. Thus (4.6) prescribes the exact root values y_j . The even residues supply every branch guard; the final residue gives the exit congruence. The threshold ensures expansion and all the inequalities in (4.1). Removing its finite initial segment preserves density, and the strictly increasing parameter-to-start map gives infinitude above every bound. $\square$

The half-open endpoints, including residue zero, are included in the formal frequency proofs. Fourier convergence yields weak convergence of the empirical measures to Haar measure. The boundaries of these boxes lie in finitely many coordinate endpoint sets of Haar measure zero, so their frequencies equal their volumes. This strengthens the open-box recurrence argument without assuming a new analytic estimate. Both corollaries fix a, b, M , the residues, and any $\varepsilon > 0$ before taking limits. They provide neither a bound on the first witness nor estimates uniform in growing parameters, and do not concatenate the finite prefixes into an infinite orbit.

4.1. Effective returns for the word OOE

The preceding results fix the modulus before taking a limit. The next result retains its dependence explicitly, for one fixed word. For integers $M, T \geq 1$, put $s_t = 1 + 2Mt$ and

$$A_M(T) = \#\{0 \leq t < T : s_t \geq 16, \quad \lfloor s_t^{9/2} \rfloor \equiv 0 \pmod{2}, \quad \lfloor s_t^{9/4} \rfloor \equiv 1 \pmod{2M}\}.$$

Theorem 4.4 (effective OOE returns). For every $M, T \geq 1$,

$$\begin{aligned} \left| A_M(T) - \frac{T}{4M} \right| &\leq \left[5 + 128M^{1/4} \left(3 + \frac{\log T}{16} \right)^2 \right] T^{63/64} + 8 \\ &\leq 2^{14} M^{1/4} T^{127/128}. \end{aligned} \quad (4.7)$$

In particular, some $0 \leq t < 2^{2176} M^{160}$ gives a start $n = s_t^2 < 2^{4354} M^{322}$ whose actual three-step itinerary is OOE, whose intermediate states are at least n , and whose exit is greater than n and congruent to $n \equiv 1 \pmod{2M}$. At $T = 2^{2176} M^{160}$, there are at least $T/(8M)$ such parameters.

Proof. Appendix C proves the uniform Fourier estimates, the finite half-open box bound, their dyadic assembly, and the witness extraction. The box is $[0, 1/2) \times [1/(2M), 1/M)$; its area is $1/(4M)$, including when $M = 1$. The exact construction of Theorem 4.1 with $a = 2, b = 1$ converts these parameter conditions into the asserted orbit conditions. $\square$

The constants are theoretical bounds, not practical search budgets. This OOE word has periodic-word denominator one; Theorem 4.4 does not make the large-denominator family of Theorem 4.1 effective. Both the counting estimate and the actual bounded return are kernel-checked. Independent mathematical review remains outstanding.

5. A signed ancestor-count theorem

Define

$$\pi_a^-(X) = \#\{1 \leq n \leq X : \exists j \geq 0, g^j(n) = a\}.$$

Let $N(a, X)$ additionally require $g^i(n) \leq X$ for every $0 \leq i \leq j$. Then $N(a, X) \leq \pi_a^-(X)$; both counts are nondecreasing in X . They count starts, not paths.

Theorem 5.1 (positive $3n - 1$ ancestors). For every positive integer a with $3 \nmid a$, there is $X_0(a)$ such that, for every integer $X \geq X_0(a)$,

$$X^{21} \leq N(a, X)^{25} \leq (\pi_a^-(X))^{25}. \quad (5.1)$$

In particular $\pi_a^-(X) \geq X^{21/25}$.

5.1. Why height changes with the sign

A fertile g -target has $a \equiv 1 \pmod{3}$ and odd predecessor $b = (2a + 1)/3 > 2a/3$. For C_{+1} , the odd predecessor at $a \equiv 2 \pmod{3}$ is $(2a - 1)/3 < 2a/3$. The homogeneous child budget $(X/a)(3/2)b$ therefore exceeds X on the minus side. For $a = 19, b = 13, X = 103$, the nominal budget is $4017/38 > 105$, admitting 104, 52, 26, 13 even though its start already exceeds the actual cutoff.

5.2. Valid grid recurrences

Let $r_0, \dots, r_{49}$ be the integer table in Appendix A and set

$$C(t) = 2^{\lfloor t/50 \rfloor} r_{t \bmod 50}, \quad F_a(t) = N(a, C(t)a).$$

Thus $C(t + 100) = 4C(t)$ and $C(50j) = 10000 \cdot 2^j$. Integer checks on the fifty residues of t give

$$8193C(t + 29) \leq 12288C(t), \quad 16386C(t) \leq 12288C(t + 21). \quad (5.2)$$

If $a \geq 4096$ and $3b = 2a + 1$, then $12288b \leq 8193a$. Combining these inequalities puts every selected child cutoff below its parent cutoff.

Lemma 5.2 (actual inverse trees). If $a \geq 4096$, $a \equiv 1 \pmod{3}$, and a is nonperiodic, then

$$\begin{aligned} F_a(t+100) &\geq F_{4a}(t), \\ F_a(t+100) &\geq F_{4a}(t) + F_b(t+129), \\ F_a(t+100) &\geq F_{4a}(t) + F_{2b}(t+79). \end{aligned} \tag{5.3}$$

The first inequality needs neither nonperiodicity nor the threshold.

Proof. The child paths are $4a, 2a, a, b, a$, and $2b, b, a$. The cutoff comparisons permit extension to a . The $4a$ subtree is disjoint from each alternative odd subtree. Otherwise a common start reaches the nonperiodic root at two different times, or the last steps force odd b to equal even $2a$. A nonperiodic root has a unique hitting time: two times would give it a positive return. Thus the indicated cardinalities add. $\square$

Use the second recurrence at $a \equiv 1 \pmod{9}$, the third at $a \equiv 7 \pmod{9}$, and only the first at $a \equiv 4 \pmod{9}$. These choices keep children in residue 1 $\pmod{3}$. The two odd alternatives are never added together.

Set $\mathcal{M}(t, a) = 10t + \lfloor \log_2(a^{498}) \rfloor$. The exact inequality

$$2^{291} 8193^{498} < 12288^{498} \tag{5.4}$$

forces the odd child's root-size term to fall by at least 291. Doubling or quadrupling adds exactly 498 or 996. Therefore

$$\begin{aligned} \mathcal{M}(t, 4a) + 4 &\leq \mathcal{M}(t+100, a), \\ \mathcal{M}(t+79, 2b) + 3 &\leq \mathcal{M}(t+100, a), \\ \mathcal{M}(t+129, b) + 1 &\leq \mathcal{M}(t+100, a). \end{aligned} \tag{5.5}$$

Even the advanced-index call decreases this natural-number measure.

5.3. A closed domain for arbitrary targets

Lemma 5.3 (root domain). Every positive target a prime to three has a nonperiodic ancestor $r > 2^{19}$ with $r \equiv 1 \pmod{3}$. All fertile ancestors of r are at least 4096 and form a domain closed under the selected productions in (5.3). For all sufficiently large X , $N(r, X) \leq N(a, X)$.

Proof. At most three doublings reach a value $z \equiv 1$ or $7 \pmod{9}$. Its two predecessors $2z$ and $(2z+1)/3$ are distinct, positive, and prime to three. They cannot both be periodic, since a deterministic map is injective on its periodic points. Select a nonperiodic predecessor, double once if necessary to reach residue 1 modulo three, then multiply by 2^{20} . This gives r as required. Ancestors of nonperiodic points are nonperiodic.

The finite barrier used here is

$$0 \leq u < 4096 \implies g^k(u) < 2^{19} \quad (k \geq 0), \tag{5.6}$$

where $g(0) = 0$. A finite descent certificate and strong induction prove it: each start reaches a smaller value or the explicit forward-invariant set of zero and the fifteen members of the three known positive cycles. Appendix A specifies an independent finite check. No exhaustiveness of the known cycles is assumed. An ancestor below 4096 could not reach $r > 2^{19}$, by (5.6). The residue calculations after (5.3) give closure. Above the maximum on the fixed path from r to a , every capped path to r extends to a capped path to a . $\square$

A threshold alone is insufficient: 4096 has odd predecessor 2731. Restricting to ancestors of r repairs that boundary.

5.4. Certificate and growth induction

Take $p = 5059$, $q = 5000$, $C_{\max} = 10^{12}$, $\mu = p/q$. The certificate has a positive integer weight c_m for each of the $3^{11} = 177147$ fertile classes modulo 3^{12} . Its minimum is 7307142888 and its maximum is $C_{\max}$. For a child class modulo 3^{11} , let $\bar{c}$ be the minimum over its three lifts modulo 3^{12} . Every row satisfies the exact integer inequality

$$\begin{aligned} c_m p^{100} &\leq c_{4m} q^{100} & (m \equiv 4 \pmod{9}), \\ c_m p^{100} &\leq c_{4m} q^{100} + \bar{c}_{(4m+2)/3} p^{79} q^{21} & (m \equiv 7 \pmod{9}), \\ c_m p^{100} q^{29} &\leq c_{4m} q^{129} + \bar{c}_{(2m+1)/3} p^{129} & (m \equiv 1 \pmod{9}). \end{aligned} \quad (5.7)$$

Indices are reduced at the indicated moduli. A minimum ensures validity for the actual child's lift, irrespective of root size.

Lemma 5.4 (growth). Throughout the domain of Lemma 5.3,

$$F_a(t) \geq \frac{c_a}{C_{\max}} \mu^{t-100} \quad (t \geq 0). \quad (5.8)$$

Proof. Induct on $\mathcal{M}(t, a)$. For $t < 100$ the root itself is counted and the right side is at most one. For $t = s + 100$, use (5.3), the decreases (5.5), and the inductive bounds. The three cases reduce exactly to (5.7) after division by the appropriate positive powers of p, q . The minimum weight does not exceed the actual child's weight. Equivalently the induction uses only integers: $c_a p^t q^{100} \leq F_a(t) C_{\max} q^t p^{100}$. $\square$

Proof of Theorem 5.1. Apply (5.8) at r and $t = 50j$. For all sufficiently large j , Lemma 5.3 gives

$$N(a, 10000r \cdot 2^j) \geq K \mu^{50j}$$

with fixed $K > 0$. Exact integer comparison gives

$$2^{21} 5000^{1250} < 5059^{1250}, \quad (5.9)$$

so $\mu^{50} > 2^{21/25}$. For $10000r \cdot 2^j \leq X < 10000r \cdot 2^{j+1}$, monotonicity gives the same lower bound at X . The strict exponential advantage eventually absorbs K and the fixed interpolation factor. Raise to the twenty-fifth power to obtain (5.1). $\square$

This is a lower bound on ancestors, not positive natural density or a conclusion about every orbit. No classification of all positive $3n - 1$ cycles enters its proof.

6. A common ceiling for the fixed grid

Keep shifts 100, 79, 129 fixed, but allow any finite residue level $\ell \geq 2$. Put $M = 3^{\ell-2}$ and take positive weights $c_0, \dots, c_{3M-1}$. For minus use representatives $3i + 1$; for plus use $3i + 2$. The normalized minus rows are (5.7) with rate $\mu > 0$; their coefficients are $\mu^{-100}, \mu^{-21}, \mu^{29}$. For plus the odd children are $(4m - 2)/3$ at $m \equiv 2 \pmod{9}$ and $(2m - 1)/3$ at $m \equiv 8 \pmod{9}$; the sterile parent is $m \equiv 5 \pmod{9}$. The same coefficients define its rows.

Theorem 6.1 (fixed-grid ceiling for both signs). If the rows hold for either sign at any finite level, and $0 < \mu$ with $\mu^{50} \leq 2$, then

$$\mu < 5069/5000, \quad \mu^{5000} < 2^{99}. \quad (6.1)$$

Thus $\gamma = 50 \log_2 \mu < 99/100$. In particular the harmonic rate $\mu^{50} = 2$ is impossible.

Proof. Put

$$S = \sum_{i=0}^{3M-1} c_i, \quad B = \sum_{j=0}^{M-1} \min(c_j, c_{j+M}, c_{j+2M}).$$

Then $3B \leq S$. The fourfold-parent index maps are $i \mapsto 4i + 1$ for minus and $i \mapsto 4i + 2$ for plus, modulo $3M$. The odd-child maps modulo M are $j \mapsto 2j$, $j \mapsto 4j + 3$ for minus, and $j \mapsto 2j + 1$, $j \mapsto 4j$ for plus. All are permutations. Summing the rows gives $S \leq \mu^{-100}S + (\mu^{29} + \mu^{-21})B$, hence

$$1 \leq F(\mu) := \mu^{-100} + \frac{\mu^{29} + \mu^{-21}}{3}. \quad (6.2)$$

The function F is convex on the positive reals, as each power has nonnegative second derivative. Exact arithmetic gives

$$\begin{aligned} F(5069/5000) &< 1, \quad F(507/500) < 1, \\ (5069/5000)^{50} &< 2 < (507/500)^{50}, \\ (5069/5000)^{5000} &< 2^{99}. \end{aligned} \quad (6.3)$$

Convexity makes $F < 1$ between the rational endpoints. Every $\mu \geq 5069/5000$ with $\mu^{50} \leq 2$ lies there, contradicting (6.2). The last comparison proves (6.1).

At the harmonic rate, clearing (6.2) gives $3\mu^{100} \leq 3 + \mu^{129} + \mu^{79}$. If $\mu^{50} = 2$, it forces $\mu^{79} \geq 3$ and then $2^{79} \geq 3^{50}$, contrary to $2^{79} < 3^{50}$. $\square$

This obstruction reflects the fixed rounding slack $79/50 < \log_2 3$. It applies at all finite table sizes, but not to every finer grid, different production system, or direct reciprocal-mass argument. It is not an upper bound for actual ancestor counts.

7. Two other limits of transfer

7.1. Counts and reciprocal mass

Proposition 7.1 (finite backward mass). For either shortcut sign, the set $\mathcal{R} = \{3 \cdot 2^k : k \geq 0\}$ is infinite and backward-closed on the positive integers, with $\sum_{n \in \mathcal{R}} 1/n = 2/3$.

Proof. A multiple of three has no odd predecessor: $3u + \sigma = 2m$ would imply $\sigma \equiv 0 \pmod{3}$. Its only predecessor is its double. Sum the geometric series $\frac{1}{3} \sum_{k \geq 0} 2^{-k}$. $\square$

The set is not forward-closed: $C_{+1}(3) = 5$, $g(3) = 4$. It refutes a statement for all backward-closed sets, not one restricted to complete fate classes. Juggler's even inverse block consists of the even integers in $[m^2, (m+1)^2)$. Its reciprocal mass has the uniform lower bound $m/(m+1)^2$ used in [C]. The code preserves neither these ordinary heights nor their reciprocal weights.

A lower bound $N(X) \gg X^\kappa$ with $0 < \kappa < 1$ alone cannot force divergent reciprocal mass. For example the distinct integers $\lfloor j^{1/\kappa} \rfloor$, $j \geq 1$, have count at least $\lfloor X^\kappa \rfloor$ and convergent reciprocal sum. Theorem 5.1 therefore supplies no missing harmonic estimate for a fate-class argument.

7.2. Stopping a word family

At fixed depth, fair words satisfy

$$\sum_{|w|=d} 2^{-d} = 1, \quad \sum_{|w|=d} 2^{-d} \rho(w) = \left(\frac{1}{2} \frac{1}{2} + \frac{1}{2} \frac{3}{2} \right)^d = 1. \quad (7.1)$$

Both identities persist for bounded complete prefix trees. The formal statement represents these as finite binary trees with both children at every internal node, and sums over their leaf words. Completeness alone does not suffice at unbounded depth.

Proposition 7.2 (moment loss at unbounded stopping). Let $\mathcal{C}$ consist of the minimal words whose multiplier first becomes less than one. It is prefix-free, and

$$\sum_{w \in \mathcal{C}} 2^{-|w|} = 1, \quad \sum_{w \in \mathcal{C}} 2^{-|w|} \rho(w) \leq 3/4. \quad (7.2)$$

Proof. The fair log-multiplier walk has mean increment $\frac{1}{2} \log(3/4) < 0$. The strong law makes its first strictly negative time finite almost surely. Its minimal stopping prefixes partition this probability-one event. Every stopped multiplier is less than one; E alone has mass 1/2 and multiplier 1/2, losing 1/4 relative to fair mass. The other terms cannot restore it. $\square$

The Lean proof instead uses the established survivor-count decay for completeness. It then regroups the nonnegative sums by word length to prove (7.2) directly on the type of minimal certificate words. The probabilistic proof explains the missing moment without invoking an optional-stopping equality.

8. What remains open

Integer transport on an actual Juggler cycle requires $D \mid A(w)$, or a useful restriction on its reduced denominator. Distribution of Juggler words requires estimates of integer-source multiplicities in code cylinders. Transport of fate contagion requires control of ordinary reciprocal mass. Existence of the exact code supplies none of these additional estimates.

The signed ancestor theorem leaves universal termination and the classification of all cycles open. Its grid ceiling is a restriction of the stated certificate family. The finite-mass ray does not settle harmonic divergence for a full fate class.

Future versions may add results with their hypotheses, proof status, prior-art comparison, and reproducibility record. The companion papers retain their own claims and histories.

Acknowledgments and use of AI

The author thanks the authors of the cited work and the Lean and Mathlib communities for the mathematical and formal infrastructure. M. Sharpe’s grid implementation is specifically acknowledged in Section 1.1 and the attributed source modules.

AI assistance was used in mathematical exploration, drafting, programming, and formalization. The author is responsible for the manuscript and its claims. Kernel-checked results are identified individually; AI assistance and local checking do not replace independent mathematical review.

Availability and versioning

The project is maintained at github.com/sneakyweasel/btlib. The canonical source is `docs/theory/juggler_signed_collatz_note.md`. Build with the command given in Appendix A; the update and validation guide is `docs/theory/PAPER_E_BUILD.md`. The release manifest records SHA-256 hashes of the manuscript, mathematical dependencies, build tools, and outputs. This edition makes no claim of an existing external deposit.

The manuscript uses CC BY 4.0. Software retains its repository license and third-party notices; reused grid material retains M. Sharpe’s MIT notice. The local deposit kit contains the PDF and a source-and-certificate archive, so finite proof data need not be recovered from a moving repository branch.

Appendix A. Finite data and reproduction

The cap table is indexed from zero:

```
10000 10140 10281 10425 10570 10718 10867 11019 11173 11329
11487 11647 11810 11975 12142 12311 12483 12658 12834 13013
13195 13379 13566 13755 13947 14142 14340 14540 14743 14948
15157 15369 15583 15801 16021 16245 16472 16702 16935 17171
```

17411 17654 17901 18150 18404 18661 18921 19185 19453 19725

The full table is stored in `data/research/juggler/negative_preimage_density/grid_k12_certificate.json`. Its ordering is $m = 3i + 1$, $0 \leq i < 177147$. For each row compute $4m \bmod 3^{12}$, and, when needed, the odd child modulo 3^{11} . Read the minimum over its three lifts, then check (5.7) with integers. Recorded floating-point diagnostics are ignored.

The weights were found by damped iteration of the positive production operator from all ones, normalizing the maximum and rounding at scale 10^{12} . The first successful table occurred after 62 iterations. Any table passing the exact checks would suffice; its discovery is outside the proof.

For an independent check of (5.6), start from all integers below 4096 and repeatedly add successors until no new state occurs. The closure of the positive starts has 6417 states and maximum 417718, below $2^{19} = 524288$; zero is fixed and can be added separately. The verifier checks closure and the strict barrier.

```
python tools/check_paper_e.py
python tools/generate_signed_grid_certificate.py --check
python tools/build_paper_e.py
python tools/build_paper_e.py --check
cd formal
lake build Problems.JugglerCollatzPaper
lake env lean AxiomCheckJugglerCollatzPaper.lean
```

The first command checks the integer certificate, grid, barrier, rational examples, modular-return witnesses, and exact comparisons (5.4), (5.9), and (6.3), together with the manuscript's numbered references and declaration inventory. These finite checks do not prove Theorem 4.1's infinite equidistribution assertion. The second command verifies the five generated Lean certificate modules against the data.

Appendix B. Formalization and review boundary

The paper barrel is `Problems.JugglerCollatzPaper`. It imports the existing proofs without changing their hypotheses.

Result	Module and declaration
Theorem 2.1	<code>CollatzPadic.orbit_bridge</code>
Series (2.4)	<code>PaperECompletion.code_hasSum_odd_times</code>
Codes and cylinders	<code>CollatzPadic.code_eq_iff</code> , <code>code_cylinder_eq</code>
Corollary 2.3	<code>PaperECompletion.all_frequency_limits_iff</code>
Example 2.2, full table	<code>PaperECompletion.example22_code_table</code>
Terminating rational codes	<code>PaperECompletion.terminating_code_positive_ternary_denominator</code>
Lemma 3.1	<code>CollatzPadic.iterate_le_of_code_eq</code>
Theorem 3.2	<code>CollatzPadic.periodic_bridge</code>
Theorem 4.1, box to orbit	<code>PaperEModularReturn.modular_return_of_box</code>
Denominator and growth	<code>PaperEModularReturn.runCode_den</code> , <code>runCode_den_gt</code>
Theorem 4.1, unconditional	<code>PaperERecurrence.theorem41</code> , <code>box_recurrence</code>
Mixed-power cancellation	<code>PowerPhaseAsymptotics.tendsto_distinct_noninteger_power_average</code>
Fourier-to-box recurrence	<code>FourierBoxRecurrence.exists_ge_fract_box_of_phase</code> ; <code>PowerBoxRecurrence.exists_ge_power_fract_box</code>
Half-open box frequencies	<code>FourierBoxCounting.tendsto_fract_box_count</code> ; <code>PowerBoxCounting.tendsto_power_fract_box_count</code>
Corollary 4.2	<code>PaperECorollaries.return_starts_asymptotic</code> , <code>returnStarts_actual</code> , <code>return_in_multiplicative_interval</code>
Corollary 4.3	<code>PaperECorollaries.signature_parameter_density</code> , <code>modular_return_of_signature</code> , <code>signature_returns_infinite</code>
Theorem 4.4, exact predicate and errors	<code>O0EEffectiveReturn.return_parameter_iff</code> , <code>count_error</code> , <code>error_power_bound</code>
Theorem 4.4, positive count and witness	<code>O0EEffectiveReturn.count_at_witnessCutoff</code> , <code>exists_bounded_modular_return</code>
Quantitative analytic inputs	<code>O0EEffectiveModes.normalized_mode_bound</code> ; <code>FejerBox.finite_box_discrepancy</code>
Lemma 5.2	<code>PreimageGrid.count_odd</code> , <code>count_doubled_odd</code> , <code>count_four</code>
Lemma 5.3	<code>PreimageDomain.closed_domain_for_target</code>
Lemma 5.4	<code>PreimageGrowth.growth_root</code>
Theorem 5.1	<code>PreimageCertificate12.density_21_25</code> , <code>ancestor_density_21_25</code>
Real exponent 21/25	<code>PaperECompletion.ancestor_density_real</code>
Theorem 6.1	<code>PreimageBalance.certificate_power_ceiling</code> , <code>rate_lt_of_rows</code> , <code>rate_lt_of_plus_rows</code>
Logarithmic ceiling	<code>PaperECompletion.certificate_log_ceiling</code>
Proposition 7.1	<code>BackwardMass.backward_mass_counterexample</code>
Section 7.1, sublinear count and finite mass	<code>SublinearCountingMass.sublinear_counting_finite_mass</code>
Finite complete trees	<code>PaperECompletion.full_prefix_tree_masses</code>
Proposition 7.2	<code>CollatzMoments.complete_family_moment_loss</code> , <code>PaperECompletion.stopping_word_masses</code>

`AxiomCheckJugglerCollatzPaper.lean` prints the dependencies of the 59 selected declarations. The permitted logical dependencies are `propext`, `Classical.choice`, and `Quot.sound`. No

additional logical axiom or native-evaluation trust extension belongs to this paper’s selected theorem audit. Theorem 4.1’s simultaneous-box premise is discharged by the included analytic proofs. Review of the correspondence between those statements and the prose is a separate responsibility.

Appendix C. Quantitative OOE proof

This appendix proves Theorem 4.4. Write $e(x) = \exp(2\pi i x)$, $s = 1 + 2Mt$, $u = \lfloor s^{9/2} \rfloor$, and $v = \lfloor s^{9/4} \rfloor$. The selected 59-declaration audit includes its complete counting and witness conclusions. The written route below uses [AR24]; the formal route derives its own sufficient derivative tests.

C.1 Formal proof route

For a sum of N consecutive samples, the finite third-derivative test, with derivative magnitude between λ and 4λ , gives

$$12N \max\{\lambda^{1/6}, (N/2)^{-1/2}, (N\sqrt{\lambda})^{-1/2}\}.$$

The fifth-derivative test, with ratio 6, gives

$$7N \max\{\lambda^{1/30}, (N/6)^{-1/8}, (N\sqrt{\lambda})^{-1/8}\}.$$

These are proved by finite differencing from a second-derivative estimate, for either constant derivative sign. On real dyadic intervals the support extends one lattice step beyond the last sample; the actual OOE derivative chains satisfy the bounds throughout that support. The module `OOEEffectiveModes` verifies both frequency axes and obtains the dyadic constant 32 and all-length constant 128 used below.

`FejerBox` proves the pointwise finite half-open box estimate, including saturated arcs and boundary hits. `OOEEffectiveReturn` identifies that box with the exact floor residues, chooses the rounded cutoff, removes at most eight initial parameters, absorbs the logarithm, and extracts the strict witness bound. Its separate audit covers all 23 theorems. Thus no analytic cancellation premise remains in Theorem 4.4’s Lean statement. This does not assert formalization of the precise external estimate (C.5).

C.2 Explicit analytic input

Write $e(x) = \exp(2 \cdot \pi \cdot i \cdot x)$. We use J. Arias de Reyna [AR24], Theorem 11 and Table 1, only at derivative orders $k = 3$ and $k = 5$. In notation adapted to this proof, if $\text{floor}(Y) > k$, f has continuous derivatives through order k on $(X, X+Y]$, and its k -th derivative lies between positive λ and Λ , then, with $D = 2^k$,

$$\frac{1}{Y} \left| \sum_{X < m \leq X+Y} e(f(m)) \right| \leq 11 \max \left\{ \left(\frac{\Lambda}{\lambda Y} \right)^{2/D}, \left(\frac{\Lambda^2}{\lambda} \right)^{1/(D-2)}, (\lambda Y^k)^{-2/D} \right\}. \quad (\text{C.5})$$

The source gives constants smaller than 11 for both orders. A derivative of constant negative sign is handled by replacing f by $-f$, which conjugates the exponential sum. No monotonicity hypothesis is needed in this higher-derivative estimate. This cited result is a written proof input, not an imported Lean theorem or a new claim of this project. Our phases are smooth for $x > 0$, and every retained interval has $Y = N \geq 6$, so $\text{floor}(Y) > 5 \geq k$. The normalization is by the real interval length Y , not by the number of integer summands.

C.3 Uniform estimates for every truncated Fourier mode

Let $H \geq 1$ be an integer and let integers h_1, h_2 satisfy $0 < \max(|h_1|, |h_2|) \leq H$. Consider

$$f_h(x) = \frac{h_1}{2}(1 + 2Mx)^{9/2} + \frac{h_2}{2M}(1 + 2Mx)^{9/4}. \quad (\text{C.6})$$

On a real interval $N < x \leq 2N$ with $N \geq \max(H^2, 6)$,

$$2MN \leq 1 + 2Mx \leq 5MN. \quad (\text{C.7})$$

Nonzero high-power coefficient

If h_1 is nonzero, the fifth derivative is

$$f_h^{(5)}(x) = \frac{945}{2}h_1M^5s^{-1/2} + \frac{945}{64}h_2M^4s^{-11/4}, \quad s = 1 + 2Mx.$$

The absolute ratio of the second term to the first is at most $H \cdot s^{-9/4}/(32M) \leq 1/32$, using $|h_1| \geq 1$, $N \geq H$, and $s \geq 2MN$. The derivative therefore has the sign of h_1 throughout the interval. The weaker factors $1/2$ and $3/2$ around its leading term give valid bounds

$$\lambda = 100|h_1|M^{9/2}N^{-1/2}, \quad \Lambda = 600|h_1|M^{9/2}N^{-1/2}. \quad (\text{C.8})$$

Indeed $945/(4 \cdot \sqrt{5}) > 100$ and $2835/(4 \cdot \sqrt{2}) < 600$. Substitution into (C.5), with $k = 5$, gives

$$\frac{1}{N} \left| \sum_{N < m \leq 2N} e(f_h(m)) \right| \leq 32H^{1/30}M^{3/20}N^{-1/60}. \quad (\text{C.9})$$

For clarity, the three terms before enlargement are bounded by $11 \cdot (6/N)^{1/16}$, $11 \cdot (3600 \cdot H \cdot M^{9/2} \cdot N^{-1/2})^{1/30}$, and $11 \cdot (100 \cdot M^{9/2} \cdot N^{9/2})^{-1/16}$. The scalar inequalities $6 < 2^1 6$ and $3600 < 2^3 0$ suffice for the constant 32.

Vanishing high-power coefficient

If $h_1 = 0$, then h_2 is nonzero and

$$f_h^{(3)}(x) = \frac{45}{16}h_2M^2s^{-3/4}.$$

Use

$$\lambda = \frac{1}{2}|h_2|M^{5/4}N^{-3/4}, \quad \Lambda = 2|h_2|M^{5/4}N^{-3/4}.$$

The endpoint comparisons follow from $45/(16 \cdot 5^{3/4}) > 1/2$ and $45/(16 \cdot 2^{3/4}) < 2$. Estimate (C.5), now with $k = 3$, bounds the normalized sum by $22 \cdot H^{1/6} \cdot M^{5/24} \cdot N^{-1/8}$. The three terms before enlargement are at most $11 \cdot (4/N)^{1/4}$, $11 \cdot (8 \cdot H \cdot M^{5/4} \cdot N^{-3/4})^{1/6}$, and $11 \cdot ((1/2) \cdot M^{5/4} \cdot N^{9/4})^{-1/4}$. Each is bounded by the displayed common expression using $H, M, N \geq 1$. Since $H^2 \leq N$,

$$H^{1/6}N^{-1/8} \leq H^{1/30}N^{-7/120} \leq H^{1/30}N^{-1/60}.$$

Together with (C.9), this proves the single bound

$$\frac{1}{N} \left| \sum_{N < m \leq 2N} e(f_h(m)) \right| \leq 32H^{1/30}M^{1/4}N^{-1/60}. \quad (\text{C.10})$$

All signs and both coordinate-axis cases are covered.

Passage to an initial segment

Suppose $1 \leq H \leq T^{1/4}$. Split $(0, T]$ into intervals $(N, 2N]$ with $N = T/2, T/4, \dots$ while $N \geq \max(H^2, 6)$. The remaining initial interval is $(0, R]$, where $R < 2 \cdot \max(H^2, 6)$; take $R = T$ if no interval is retained. The intervals are disjoint as half-open sets, even at noninteger endpoints. The remainder has $\text{floor}(R)$ integer summands. The change from indices $1, \dots, T$ to $0, \dots, T-1$ is exactly $e(f_h(0)) - e(f_h(T))$, of norm at most 2. Their total contribution is at most $\text{floor}(R) + 2 \leq 2H^2 + 14 \leq 16H^2$, including when no interval is retained. Moreover,

$$\sum_{j \geq 0} (T/2^{j+1})^{59/60} = \frac{T^{59/60}}{2^{59/60} - 1} < 2T^{59/60}.$$

Since $H^2 \leq T^{1/2}$, (C.10) and the initial remainder imply

$$\frac{1}{T} \left| \sum_{0 \leq t < T} e(f_h(t)) \right| \leq 128M^{1/4}H^{1/30}T^{-1/60}. \quad (\text{C.11})$$

The intermediate constant is at most $64 + 16 = 80$; 128 is a convenient enlargement. Integer endpoints require no integrality of the dyadic N .

C.4 An explicit half-open box estimate

For arbitrary points $z_0, \dots, z_{T-1}$ on the two-dimensional unit torus, let E_H bound the normalized sums of every nonzero Fourier mode with $\max(|h_1|, |h_2|) \leq H$. For every half-open product of circular intervals B ,

$$\left| \frac{\#\{t < T : z_t \in B\}}{T} - |B| \right| \leq \frac{5}{\sqrt{H+1}} + (3 + 2 \log H)^2 E_H. \quad (\text{C.12})$$

Here is a proof with constants and endpoints retained. The Fejer kernel

$$F_H(x) = \frac{1}{H+1} \left| \sum_{j=0}^H e(jx) \right|^2$$

is nonnegative, has integral one, and has Fourier coefficients $1-|h|/(H+1)$ for $|h| \leq H$ and zero otherwise. For $0 < \delta \leq 1/2$, $\sin(\pi \cdot x) \geq 2 \cdot x$ on $0 \leq x \leq 1/2$ gives

$$\int_{\|x\| \geq \delta} F_H(x) dx \leq \frac{1}{2(H+1)\delta}.$$

The product kernel $K = F_H$ tensor F_H therefore has mass at most $q = 1/((H+1) \cdot \delta)$ outside the coordinate delta-neighbourhood of zero. Expand and contract each interval of B by delta on the circle, obtaining B^+ and B^- . A contracted interval can be empty and an expanded one full; both cases are retained. More precisely, write a proper circular interval as the image of $[a, a+\text{ell})$, with $0 < \text{ell} < 1$. Its expansion is the image of $[a-\text{delta}, a+\text{ell}+\text{delta})$ if $\text{ell} + 2 \cdot \delta < 1$, and the whole circle otherwise. Its contraction is the image of $[a+\text{delta}, a+\text{ell}-\text{delta})$ if $\text{ell} > 2 \cdot \delta$, and empty otherwise. Empty and full original intervals are left unchanged. For shifts y with each circular distance $\|y_i\| < \delta$, these definitions give z in B implies $z-y$ in B^+ , and $z-y$ in B^- implies z in B . The exceptional shifts have K -mass at most q . Since $0 \leq \mathbf{1}_B \cdot K \leq 1$, the following inequalities hold at every point, including the original interval endpoints:

$$\mathbf{1}_{B^-} * K - q \leq \mathbf{1}_B \leq \mathbf{1}_{B^+} * K + q.$$

The changes in volume are at most $4 \cdot \delta$. A circular interval's Fourier coefficient has absolute value at most $1/\max(1, |h|)$, also for the empty or full interval. Thus the sum of absolute nonconstant coefficients of either convolution is at most

$$\left(1 + 2 \sum_{h=1}^H \frac{1}{h}\right)^2 - 1 \leq (3 + 2 \log H)^2.$$

Averaging the pointwise inequalities proves an error at most $4 \cdot \delta + q + (3 + 2 \cdot \log(H))^2 \cdot E_H$. Take $\delta = (H + 1)^{-1/2}$ for $H \geq 3$. For $H = 1$ or 2 , the trivial discrepancy bound one already implies (C.12).

C.5 Assembly and first-witness extraction

Apply (C.12) to

$$z_t = \left(\frac{s^{9/2}}{2}, \frac{s^{9/4}}{2M} \right) \pmod{1}, \quad B = [0, 1/2) \times [1/(2M), 1/M).$$

Its area is $1/(4M)$. For any $x \geq 0$ and integers $m \geq 1$ and $0 \leq r < m$, writing $\text{floor}(x) = m \cdot q + r'$ with $0 \leq r' < m$ gives $\text{fract}(x/m) = (r' + \text{fract}(x))/m$. Consequently $\text{floor}(x) = r$ modulo m is equivalent to $r/m \leq \text{fract}(x/m) < (r+1)/m$, including equality at the left endpoint. Box membership is therefore exactly u even and $v = 1$ modulo $2M$. For $M = 1$ the second interval is $[1/2, 1)$: its right endpoint is excluded; an integer value of $x/(2M)$ has fractional part zero and is rejected. The first interval likewise excludes its endpoint $1/2$. No absence of boundary hits is assumed. Choose $H = \text{floor}(T^{1/32})$, so $H \geq 1$, $H \leq T^{1/4}$, and

$$(H + 1)^{-1/2} \leq T^{-1/64}, \quad H^{1/30} T^{-1/60} \leq T^{-1/64}, \quad 2 \log H \leq \frac{\log T}{16}.$$

Equations (C.11) and (C.12) give the first bound in (4.7) before the final $+8$. Exactly $\min(T, 1 + \text{floor}(7/M))$ parameters have $1 + 2Mt < 16$. Deleting the box hits among these parameters changes the count by at most eight.

For the second bound in (4.7), put $x = \log(T) \geq 0$. The maximum of

$$(3 + x/16)^2 e^{-x/128}$$

on this half-line is $256 \cdot \exp(-13/8)$, attained at $x = 208$, and is less than 64. For example, the first four terms of the exponential series already give $\exp(13/8) > 4$. Absorb $8/T$ into $8 \cdot T^{-1/128}$ and $5 \cdot T^{-1/64}$ into $5 \cdot T^{-1/128}$. Since $M \geq 1$ and $5 + 128 \cdot 64 + 8 = 8205 < 2^{14}$, the second inequality follows.

At $T = 2^{2176} M^{160}$, one has $T^{1/128} = 2^{17} \cdot M^{5/4}$. Hence the error is at most $T/(8M)$, while the main term is $T/(4M)$. Thus $A_M(T) \geq T/(8M) > 0$. Select a counted t . Since $t < T$ is integral, $1 + 2Mt < 2MT$, giving the parameter and start bounds in Theorem 4.4.

The actual orbit is

$$s^2 \mapsto s^3 \mapsto u \mapsto v.$$

Both initial sources are odd, the third is even by construction, and the exact identity $\text{isqrt}(\text{isqrt}(s^9)) = \text{floor}(s^{9/4})$ identifies the exit. The already proved size threshold $s \geq 16$ gives $v > s^2$; the first two images are larger still. Finally $s = 1$ modulo $2M$ gives $n = 1$ modulo $2M$, and the box gives the exit congruence in Theorem 4.4. This completes the written proof.

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